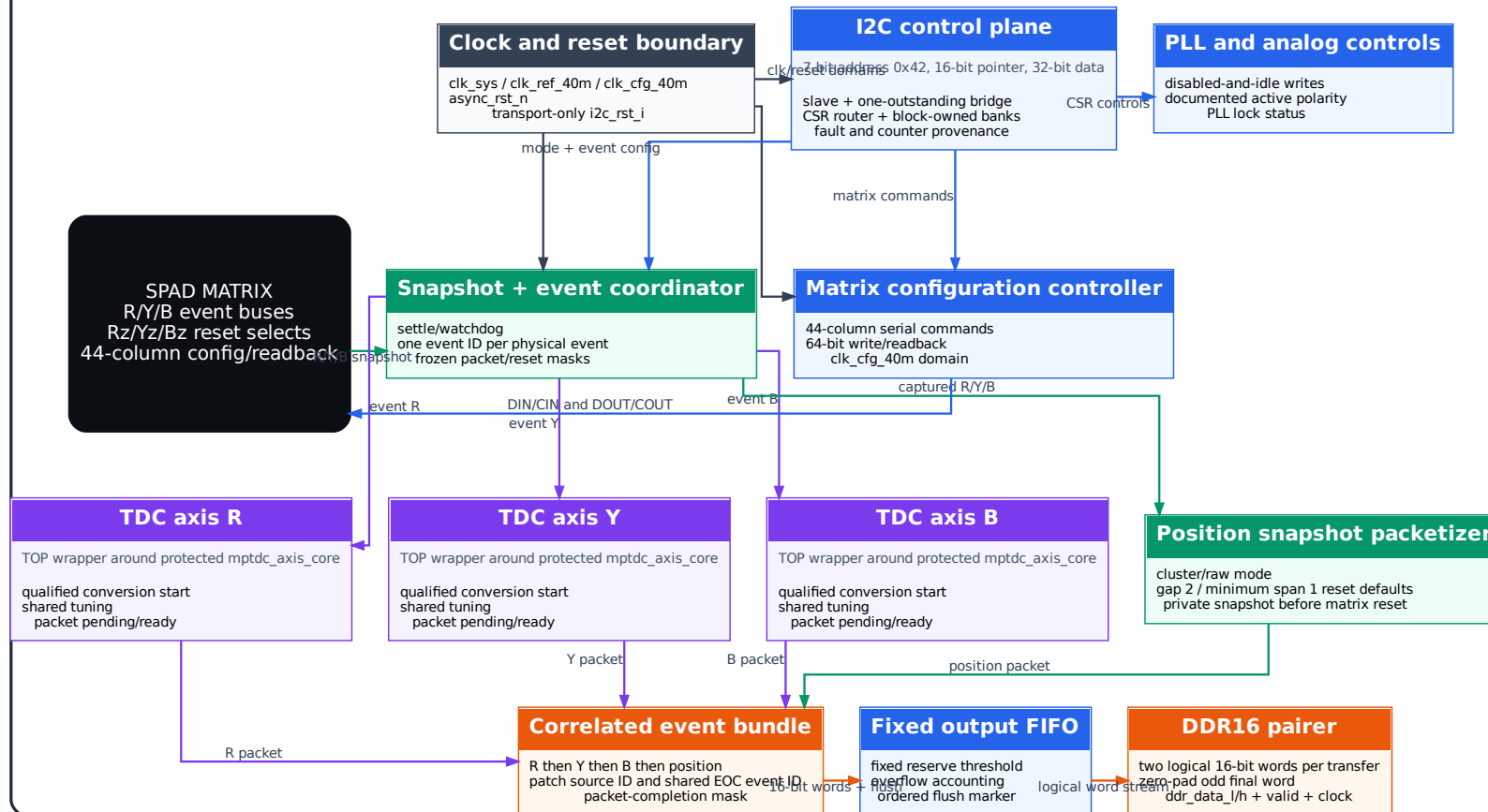


SPADMIC active matrix-top architecture

Current CSR/I2C, R/Y/B event, correlated packet, matrix configuration, and DDR16 data paths.
TOP/rtl/spadmic_top_matrix_v1.sv + TOP/docs/01_ACTIVE_ARCHITECTURE.md

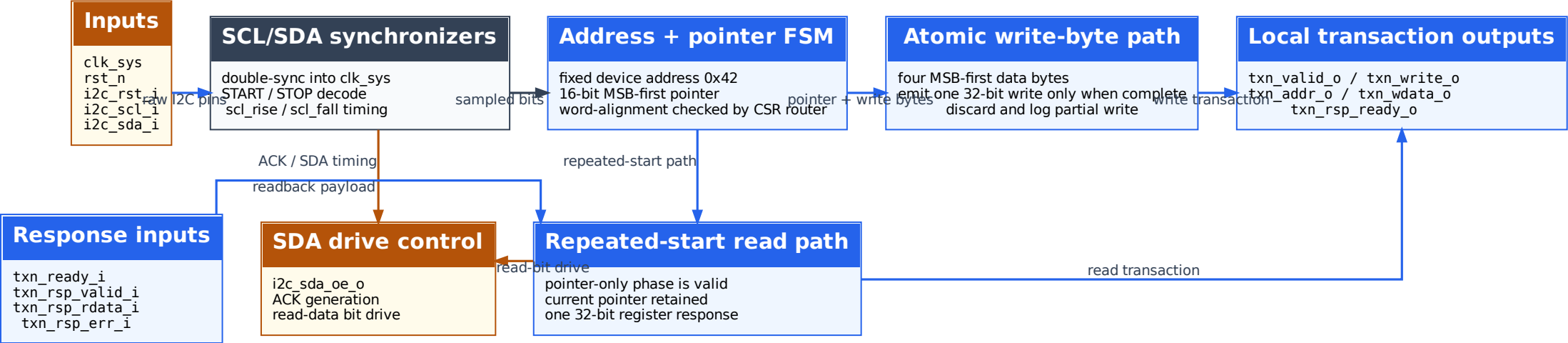
SPADMIC active matrix-top architecture



spadmic_i2c_slave

Synchronized I2C front end with atomic 32-bit transactions, partial-write discard, and transport-only reset.

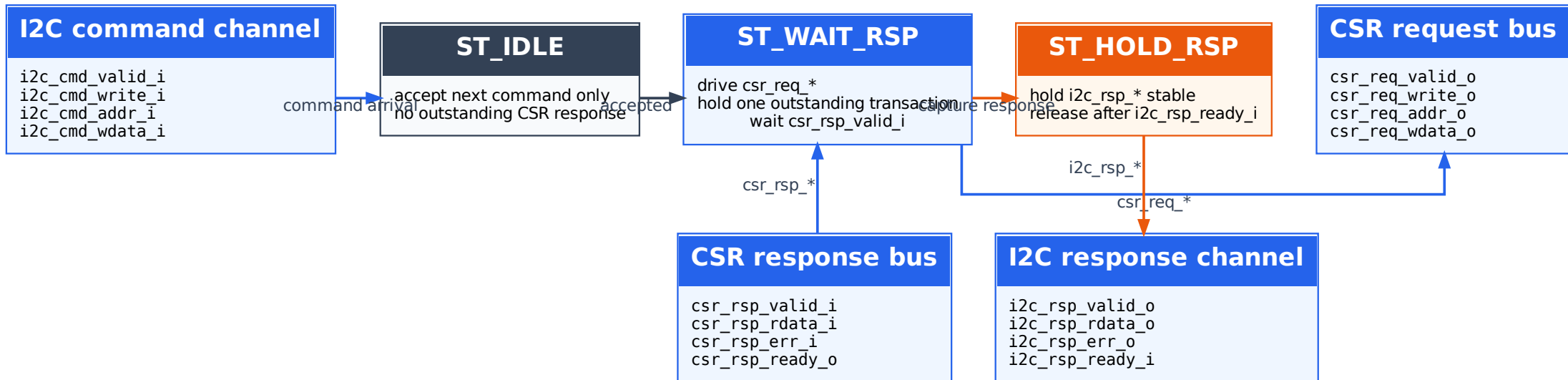
I2C/rtl/spadmic_i2c_slave.sv



spadmic_i2c_csr_bridge

Single-outstanding bridge between the I2C transaction channel and the shared local CSR bus.

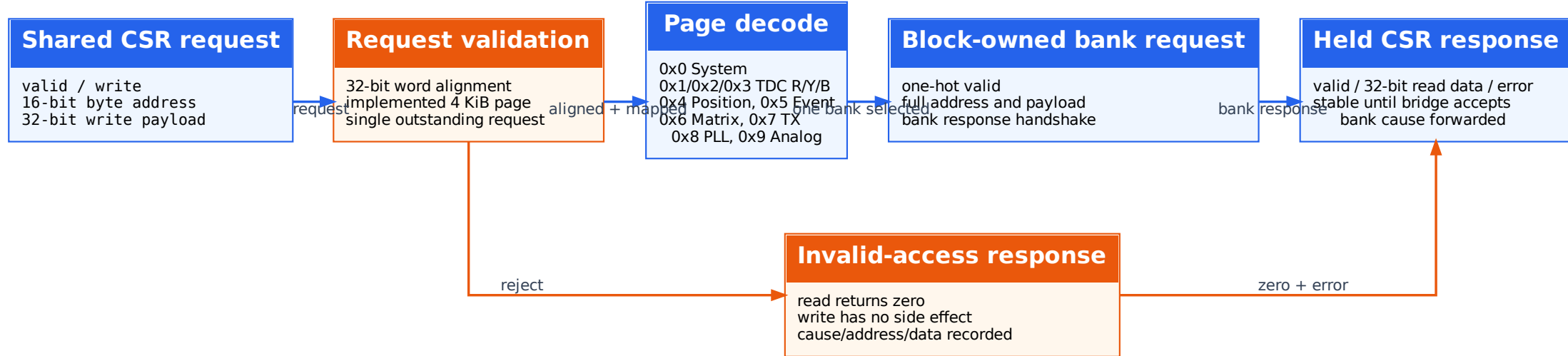
I2C/rtl/spadmic_i2c_csr_bridge.sv



spadmic_csr_router

ABI 1.0 request validation, page routing, and fail-closed invalid-access behavior.

TOP/rtl/spadmic_csr_router.sv



SPADMIC block-owned CSR banks

ABI 1.0 register ownership, write admission, W1C fault handling, and saturating diagnostics.

TOP/rtl/spadmic_csr_banks.sv + TOP/rtl/spadmic_csr_map_pkg.sv

System bank

ID and ABI version
atomic GLOBAL_CTRL
safe-idle and access diagnostics
maintenance counter clear

TDC R/Y/B banks

per-axis status
W1C FIFO fault
saturating error count

Position bank

cluster/raw mode
gap and minimum span
drop fault/count

Event bank

event lifecycle and masks
snapshot/reset configuration
R/Y/B snapshot readback

Matrix bank

column command and 64-bit data
readback and validity
W1C command fault

TX bank

bundle/FIFO/DDR status
fixed FIFO geometry
missing/overflow faults and counts

PLL bank

PLL control image
lock status
disabled-and-idle write gate

Analog bank

SLVS/RX controls
explicit active polarity
disabled-and-idle write gate

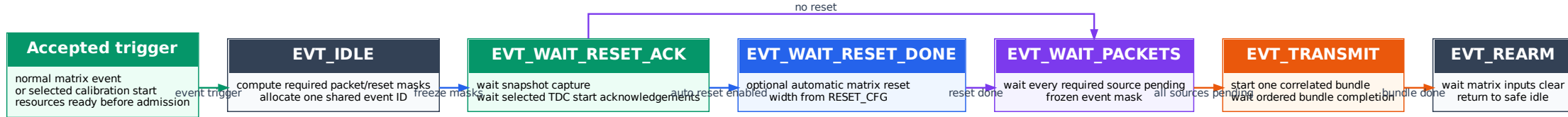
Shared bank policy

Configuration writes require global disabled and safe idle.
GLOBAL_CTRL validates and commits a complete image atomically.
Faults are W1C; maintenance clears counters only while disabled and idle.

spadmic_event_coordinator

One-event-at-a-time lifecycle with frozen source masks, reset acknowledgements, and one shared event ID.

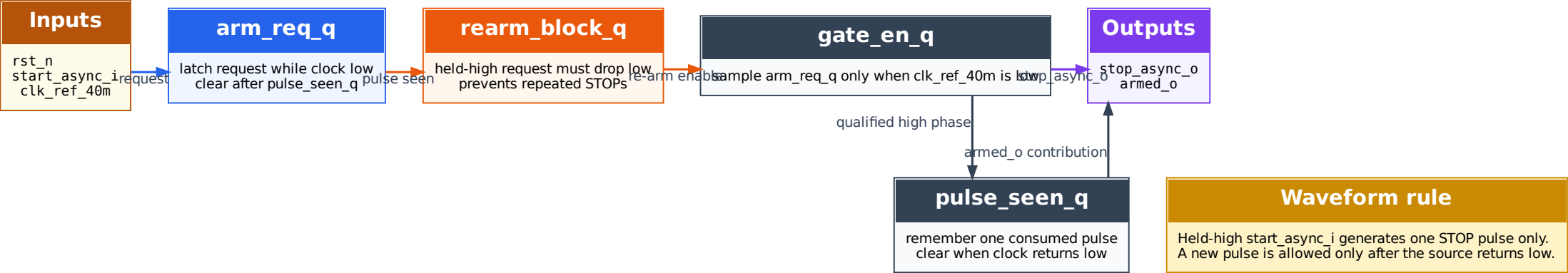
TOP/rtl/spadmic_event_coordinator.sv



spadmic_ref_stop_qualifier

One-shot STOP qualifier with explicit held-high re-arm blocking.

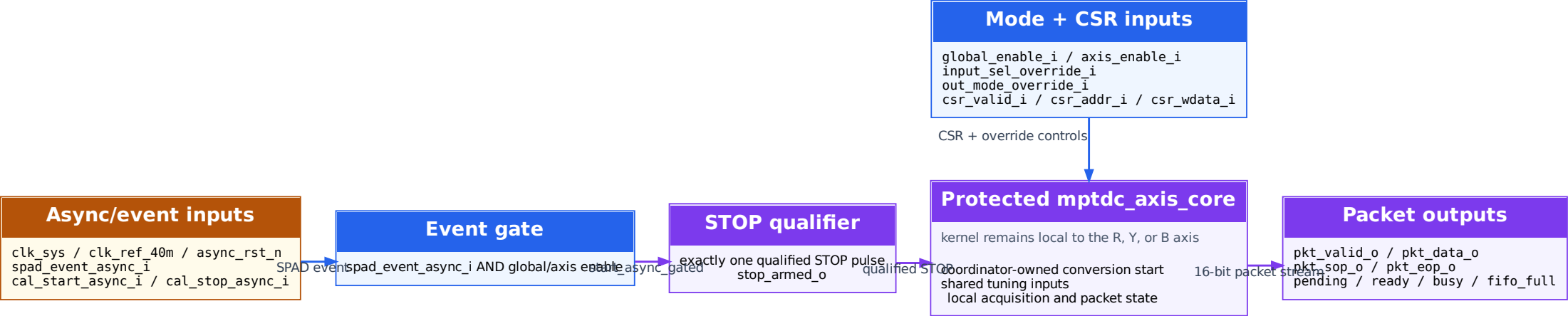
TOP/rtl/spadmic_ref_stop_qualifier.sv



spadmic_tdc_axis_wrapper

TOP-owned R/Y/B wrapper around the stop qualifier and protected MPTDC axis core.

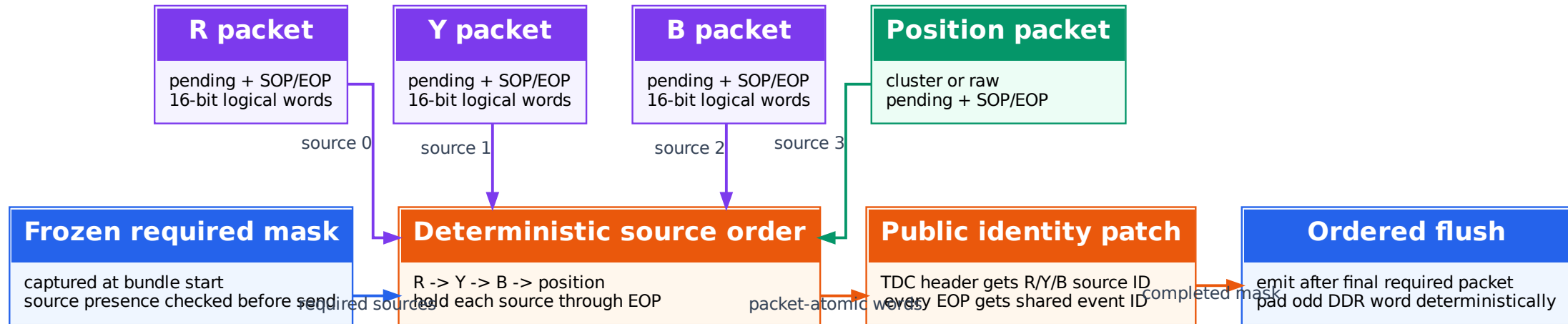
TOP/rtl/spadmic_tdc_axis_wrapper.sv



spadmic_event_bundle_tx

Deterministic R/Y/B/position packet bundle with one shared event ID and an ordered flush marker.

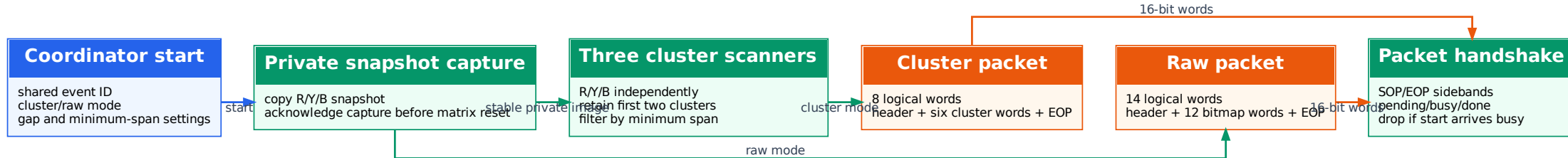
TOP/rtl/spadmic_event_bundle_tx.sv



spadmic_position_snapshot_packetizer

Private R/Y/B snapshot capture with cluster/raw packet generation under coordinator control.

TOP/rtl/spadmic_position_snapshot_packetizer.sv



spadmic_axis_cluster_scan

Two-cluster bitmap scanner with threshold-based split and overflow retention of the first two clusters.

position/rtl/spadmic_axis_cluster_scan.sv

